

**CISS362: Introduction to Automata Theory, Languages, and
Computation
Assignment a10**

Name: _____

OBJECTIVES

- Design context-free grammars.
- Design PDA.
- Prove a language is not context-free.

For questions asking you show an operation on context-free languages language is closed, you need not prove your construction is correct. (But you are welcome to do so. Most of such proofs, if not immediate, is by induction.) If the question does not say so, you can choose to give a PDA or a context-free grammar. For a PDA, draw the PDA state diagram and give the formal definition.

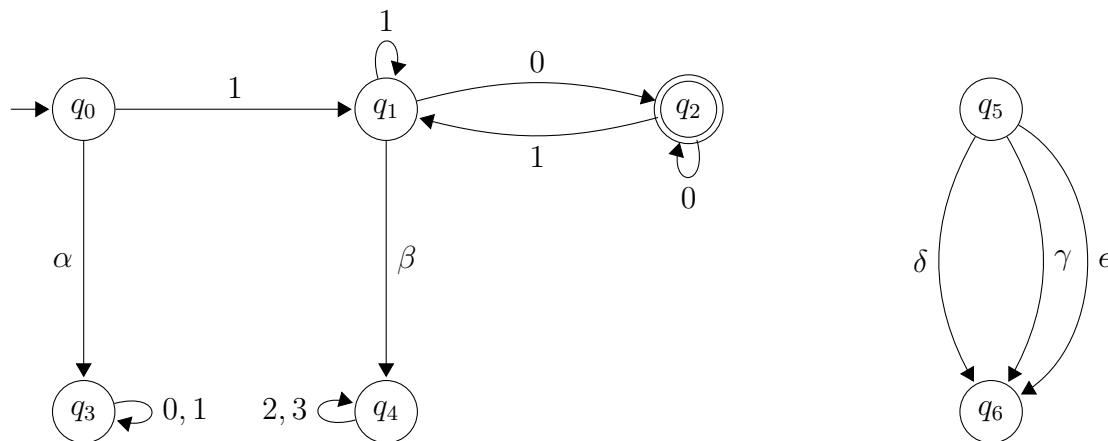
Study the solutions to the following questions in the textbook: 2.18, 2.30(b) and (c)

Here are the questions you should work. Some questions have solutions (either from the book or I have written up the solution).

- Sipser 2.11. Q1.
- Sipser 2.12. Q2.
- Sipser 2.14. Q3.
- Sipser 2.15. Solution to 2.15 is provided.
- Sipser 2.16. Q4.
- Sipser 2.17. Solution to 2.17 is provided.
- Sipser 2.18. I talked about this in class. Solution to 2.18 is also provided in the textbook – study it carefully.
- Sipser 2.19. Q5. Solutions to 2.6(a) and 2.6(c) are provided in the textbook. Study it carefully.
- Sipser 2.20. Q6.
- Sipser 2.30. Q7.
- Sipser 2.31. Q8.
- Sipser 2.32. Q9.
- Q10.

HOW TO DRAW A STATE DIAGRAM

Here's an example showing you how to draw the elements of a state diagram. Also, look at the solution to 1.3 below.



For more information on drawing state diagrams go to my tutorials and look for `latex-automata.pdf`:

<https://drive.google.com/file/d/1AeE-POWNvQlitzPDxQpGE8bMR9Yc9gMW>

Let me know if you have any questions about drawing state diagram.

HOW TO WRITE A CONTEXT-FREE GRAMMAR

Here's a context-free grammar, G , for our $\{a^n b^n \mid n \geq 0\}$:

$$G : \begin{cases} S \rightarrow \epsilon \\ S \rightarrow aSb \end{cases}$$

Usually we combine the productions for the same variable like this:

$$G : S \rightarrow \epsilon \mid aSb$$

Here's an example of a longer grammar:

$$G_1 : \begin{cases} S \rightarrow \epsilon \mid TU \\ T \rightarrow a \mid aT \\ U \rightarrow \epsilon \mid bUcc \end{cases}$$

And remember to list the starting variable first.

HOW TO DRAW A CYK TABLE

$i \backslash j$	b	a	a	a	a	b
	1	2	3	4	5	6
1	$\{B, C\}$	$\{A\}$	$\{A\}$	$\{A\}$	$\{A\}$	$\{B, C\}$
2	$\emptyset$	$\{C\}$	$\{C\}$	$\{C\}$	$\{S\}$	
3	$\{A\}$	$\{S\}$	$\{S\}$	$\{B\}$		
4	$\{C\}$	$\emptyset$	$\{S\}$			
5	$\{S\}$	$\{B\}$				
6	$\{B, S\}$					

The following shows you how to control the size of the cells:

$i \backslash j$	b	a	a	a	a	b
	1	2	3	4	5	6
1	$\{B_b, C_b\}$	$\{A_a\}$	$\{A_a\}$	$\{A_a\}$	$\{A_a\}$	$\{B_b, C_b\}$
2	$\emptyset$	$\{C_{(1,2),(1,3)}\}$	$\{C_{(1,3),(1,4)}\}$	$\{C_{(1,4),(1,5)}\}$	$\{S_{(1,5),(1,6)}\}$	
3	$\{A_{(1,1),(2,2)}\}$	$\{S_{(1,2),(2,3)}\}$	$\{S_{(1,3),(2,4)}\}$	$\{B_{(2,4),(1,6)}\}$		
4	$\{C_{(3,1),(1,4)}\}$	$\emptyset$	$\{S_{(1,3),(3,4)}\}$			
5	$\{S_{(3,1),(2,4)}\}$	$\{B_{(2,2),(3,4)}\}$				
6	$\{B_{(1,1),(2,5)}\}$ $S_{(3,1),(3,4)}\}$					

Q1. Sipser 2.11.

SOLUTION.

Q2. Sipser 2.12.

SOLUTION.

Q3. Sipser 2.14.

SOLUTION.

Sipser 2.15.

Give a counter-example to show that the following fails to prove that the class of context-free languages is closed under Kleene star.

“Let L be a CFL that is generated by the CFG $G = (\Sigma, V, S, P)$ where V is the set of variables, $S \in V$ is the starting symbol, and P is the set of production rules. Add the new rule $S \rightarrow SS$ and call the resulting grammar G' . G' generates L^* .”

SOLUTION.

Consider the context-free grammar $G : S \rightarrow a$. Then $L(G')$ does not contain ϵ . (In fact $L(G') = \{a^n \mid n \geq 2\}$.) $\square$

Q4. Sipser 2.16

Show that the class of context-free languages is closed under union, concatenation, and Kleene star.

(Note that I have actually already given you the proofs in class. You just have to write it up in proper mathematical prose.)

SOLUTION.

Sipser 2.17. Use the results of Exercise 2.16 to give another proof that every regular language is context free, by showing how to convert a regular expression directly to an equivalent context-free grammar.

SOLUTION.

We will prove by induction on the length of the regular expression r such that $L(r)$ is context-free. Let $P(n)$ be the statement that if r is a regular expression of length n , then $L(r)$ is context-free.

BASE CASE. Let r be a regular expression of length 1. Then either $r = c \in \Sigma$ or $r = \emptyset$. If $c \in \Sigma$, then $L(c) = \{c\}$ is context-free since the grammar $S \rightarrow c$ generated $L(c)$. Also, the grammar $S \rightarrow S$ generates $L(\emptyset)$. Therefore $P(1)$ holds.

INDUCTIVE CASE. Assume $P(k)$ holds for $1 \leq k \leq n$. We will show that $P(n+1)$ holds where $n+1 \geq 2$. Let r be a regular expression of length $n+1$. Then r is $(r_0), r_0^*, r_0 \cup r_1, r_0 \cdot r_1$. In all cases r_0 or r_0, r_1 are regular expressions of length $\leq n$. Hence $L(r_0)$ or $L(r_0), L(r_1)$ are context-free. Since context-free languages are closed under concatenation, union, and Kleene star, $L(r)$ is also context-free. (The case of $r = (r_0)$ is immediate and does not require closure of concatenation, union, Kleene star.) Hence $P(n+1)$ holds.

Therefore by strong mathematical induction $P(n)$ holds for all $n \geq 1$, i.e., all regular languages are context-free. $\square$

Q5. Sipser 2.19.

SOLUTION.

Q6. Sipser 2.20.

SOLUTION.

Q7. Sipser 2.21. (You need not provide the proof.)

SOLUTION.

Q8. Sipser 2.22.

SOLUTION.

Q9. Sipser 2.32.

SOLUTION.

Q10. Consider the grammar G given by

$$\begin{aligned} S &\rightarrow AB \mid BA \\ A &\rightarrow AC \mid a \\ B &\rightarrow SA \mid SB \mid b \\ C &\rightarrow SS \mid BA \mid a \end{aligned}$$

Using the grammar G above, check if the following words are generated by G using the CKY algorithm. Provide the completed CYK tables. For words generated by G , write down all their possible derivations.

- (a) aa
- (b) bab
- (c) $abab$
- (d) $aabaab$

SOLUTION.